CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 11: Areas Related to Circles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the area of a sector of a circle of radius 9 cm with angle 40°. Take $\pi = \frac{22}{7}$.
2. Find the area of a quadrant of a circle whose circumference is 44 cm.
3. The minute hand of a clock is 7 cm long. Find the area it sweeps in 15 minutes.
4. Find the length of the arc of a sector of angle 120° in a circle of radius 21 cm.
5. Find the perimeter of a sector of radius 10.5 cm and angle 60°.
6. A chord of a circle of radius 20 cm subtends a right angle at the centre. Find the area of the minor segment. Take $\pi = 3.14$.
7. For the circle in the previous question, find the area of the major sector.
8. A chord of a circle of radius 14 cm subtends 60° at the centre. Find the area of the minor segment. Take $\sqrt{3} = 1.73$.
9. A car has two wipers that do not overlap. Each wiper has a blade 21 cm long sweeping through 120°. Find the total area cleaned in one sweep.
10. An arc of a circle of radius 14 cm subtends 45° at the centre. Find the area of the sector and of the corresponding major sector.

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Answer Key

1. $\frac{198}{7} \approx 28.29$ cm²
2. $r = 7$ cm; area $= 38.5$ cm²
3. $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$ cm²
4. 44 cm
5. $11 + 21 = 32$ cm
6. $314 - 200 = 114$ cm²
7. $1256 - 314 = 942$ cm²
8. $\frac{308}{3} - 49 \times 1.73 \approx 102.67 - 84.77 = 17.9$ cm²
9. $2 \times 462 = 924$ cm²
10. 77 cm² and $616 - 77 = 539$ cm²

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